

Prakhar Shekhar Parthasarathi

Full-Stack Software Engineer · Java / Spring Boot · React · AWS

prakhar.mnnit.2022@gmail.com · Shahjahanpur, UP · open to relocate · github.com/praxstack · linkedin.com/in/prakharshekhar · prax-portfolio-one.vercel.app

47% p90_i · +33% conv. · 195k rps GET · 43 bps CPT_i · 20 → 5 on-call / shift

SUMMARY

Full-stack engineer with **3+ years at Amazon** (Travel — Hotels & Flights); owned **4 microservices** end-to-end. Backend-heavy: **Java 17 / Spring Boot**, JVM concurrency (`CompletableFuture`), bounded `ExecutorService`, idempotent event-driven flows over **SQS + Step Functions**, IaC in **AWS CDK (Java)**; front-end: **React (SSR & CSR) + Redux**. Impact: 47% p90 cut, 33% conversion lift, 43 bps CPT reduction, on-call 20+ → ~5 per shift.

CORE TECHNICAL STACK

JAVA & JVM	Java 17, Spring Boot/MVC, Guice, JUnit 5, Mockito, Maven · concurrency, GC profiling
DISTRIBUTED	Microservices, REST, GraphQL, idempotency, state machines, DLQ, at-least-once, TDD
AWS	Lambda, DynamoDB, SQS, Step Functions, API Gateway, CloudWatch, ECS, CDK (Java)
FRONT-END	React (SSR & CSR), Redux, HTML/CSS, TypeScript
DATA	DynamoDB, Redis, NoSQL modelling, SQL, caching
TOOLING	Git, CI/CD, JaCoCo gates, Agile, on-call, incident response · Python, C++

EXPERIENCE

Software Development Engineer Jul 2022 – Sep 2025

Amazon · Travel (Hotels & Flights) · Bengaluru

- Ownership. Owned **4 customer-facing microservices** (search → order review); Spring Boot REST, DynamoDB DAOs, Prime/partner discount flows — ~30 bps funnel churn drop.
- Concurrency. Parallelised 6 partner APIs via `CompletableFuture.allOf()` — p90 47% (1.1s→0.59s), conversion +33%; GZIP cut wire size 95%.
- Event-driven. Idempotent flight schedule-change webhook (SQS, dedup keys) — 43 bps CPT; Step Functions booking state machine with Java Lambdas.
- IaC & ops. All service infra in **Java CDK**; migrated 2 services from CloudFormation — rollback incidents halved. On-call tickets 20+→~5/shift; 80%+ coverage, 150+ incidents resolved.

Software Development Engineer, Intern 2021 – 2022

Amazon · Bengaluru

- Migrated GraphQL resolvers to Java + DynamoDB; shipped **One-Click Renewal** for auto-insurance; 90%+ unit-test coverage (JUnit 5 + Mockito).

SELECTED PROJECTS

AI Visual Code Review github.com/praxstack/ai-visual-code-review · npm v2.4.1 Visual diff review tool + VS Code extension; TypeScript, Jest, command-injection hardening, AI markdown exports.	Redis Server in Java github.com/praxstack/redis-server-java RESP2 server from scratch; bounded thread pool, <code>ConcurrentHashMap</code> , hybrid TTL. Benchmarks: GET 195k rps, SET 154k rps; 42 tests.
Coach Atlas github.com/praxstack/coach-atlas AI interview mentor + tutorial studio; React/TS, Mermaid/KaTeX rendering, multi-provider BYOK (keys in browser).	Open Source — Fabric (42k★) danielmiessler/Fabric 3 merged PRs: Bedrock bearer auth (#2044), dynamic regions (#2052), streaming deadlock fix (#2061).

EDUCATION & RECOGNITION

M.C.A., Computer Science · MNNIT, Prayagraj GPA 9.25	2019 – 2022
B.Sc. (Hons), Computer Science · University of Delhi	2015 – 2018
Amazon Pay Merchants Categories Champion — 100+ trust-buster bugs; on-call 20+→~5/shift	2023